



Mitosis and the cell cycle

Today I can... ☐ I can order interphase, four mitosis phases and cytokinesis. ☐ I can show that the two new body cells have matching chromosome sets. ☐ I can explain how mitosis helps growth or repair.

Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R _____

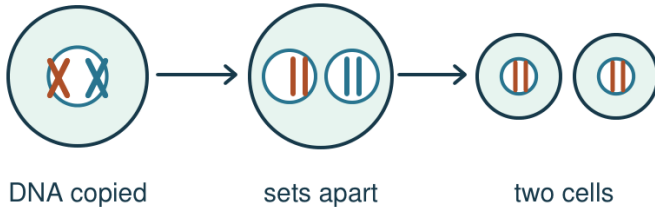
Arrival

Point to the first and last picture in the cell strip. What changed?

Interphase

Mitosis

Cytokinesis



My response

Starter

Recall three things from earlier lessons. Show one answer at a time.

1. Which process uses energy to move a substance from low to high concentration?

2. In osmosis, which substance moves through a partially permeable membrane?

3. In a eukaryotic animal cell, which structure holds the chromosomes?



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Talk-then-sort

Complete the missing step in the part-done model. Then place the paper strips in both new cells.

Interphase: copy DNA. Mitosis: prophase (condense), metaphase (line up), _____, telophase (two nuclei).

Cytokinesis: split the cell.

Choices: Interphase · Mitosis · Cytokinesis

Card or step	My group or match
First	
Then	
Last	

Use the model to explain how cell division helps replace damaged body cells.

Independent F

Order the three cards and show the same chromosome colours in each new cell.

First ...; then ...; last ...

My response

Independent M

Explain why each new cell has a matching chromosome set.

Each new cell matches because ...

My response



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Independent H

Predict what would happen to the sets if copied chromosomes did not separate evenly.

If the copies did not separate evenly, then ... because ...

My response

Go deeper

How do copy, separate and split work together when new cells are needed for repair?

My response

Fault finder: compare two faulty models. One skips DNA copying; the other puts both copies of one chromosome in one nucleus. Find each first error and propose a repair.

My response

Exit

Show how you know: order the three stages and explain why the new cells match.

My exit response

My feedback was ... Name the feedback you heard at Review.

Next I need to ... Check both matching chromosome sets and the final cell split.



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Word help

Word	Meaning	Example	My language
cell cycle	the series of steps in which a cell grows and divides	The cell cycle includes interphase, mitosis and cytokinesis.	
interphase	the stage in which the cell grows and copies its DNA	The cell copies its DNA in interphase.	
mitosis	the stage in which copied chromosomes separate into two nuclei	Mitosis separates the chromosome sets.	
chromosome	a structure of DNA that carries genes	Each new nucleus receives a matching set of chromosomes.	
cytokinesis	the division of the cell after the nuclei form	Cytokinesis splits one cell into two.	
prophase	the mitosis phase when chromosomes condense	In prophase, chromosomes become easier to see.	
metaphase	the mitosis phase when chromosomes line up in the middle	In metaphase, chromosomes form a line.	
anaphase	the mitosis phase when the two copies of each chromosome are pulled to opposite ends	In anaphase, the two copies of each chromosome are pulled to opposite ends.	
telophase	the mitosis phase when two new nuclei form	In telophase, each set gains a nucleus.	